Equity Dispersion Volatility Swap Payoff

1 Trade Structure

An equity dispersion volatility swap package is usually composed of:

- long volatility swaps on a basket of single-name equities;
- short volatility swap on the equity index, typically SPX.

The economic exposure is therefore:

$$\Pi_{\text{disp}} = \sum_{j \in \mathcal{S}} \Pi_j - \Pi_I, \tag{1}$$

where $\mathcal{S}$ is the set of single-name share transactions and I is the index transaction. The volatility amounts or vega notionals in the term sheet already include the intended trade sizing, so no additional weight is needed unless the model starts from a raw index notional and separately allocates it across names.

2 Payoff Versus Valuation

The formulas below first define the terminal payoff at maturity. A mark-to-market valuation before maturity is not the payoff itself; it is the discounted expectation of that payoff under the risk-neutral measure:

$$V_t = D(t,T)\mathbb{E}_t^{\mathbb{Q}}\left[\Pi_{\text{disp}}(T)\right], \tag{2}$$

where $D(t,T)$ is the discount factor from maturity T to valuation time t. In practice, the expectation is estimated by combining realized variance already observed with a market-implied estimate of the remaining variance. The projected payoff formulas in the interim valuation section should therefore be read as a practical approximation to this expectation.

3 Final Realized Volatility

For each underlying j, let $P_{j,i}$ be the official closing price or level on observation day i. Define the daily log return as:

$$r_{j,i} = \ln\left(\frac{P_{j,i}}{P_{j,i-1}}\right). \tag{3}$$

The term-sheet style final realized volatility is:

$$\mathrm{FRV}_j = 100\sqrt{\frac{252}{N}\sum_{i=1}^{N} r_{j,i}^2}, \tag{4}$$

where N is the expected number of scheduled trading days in the full observation period. The factor 252 annualizes the realized variance, and the multiplier 100 expresses volatility in percentage points.

4 Single Leg Payoff

For a volatility swap leg j, define:

Symbol	Meaning
A_j	Volatility amount or vega notional, in USD per volatility point
K_j	Volatility strike, in volatility points
C_j	Volatility cap, in volatility points, if applicable
FRV_j	Final realized volatility, in volatility points

If the leg is capped, the payoff to the volatility buyer is:

$$\Pi_j = A_j\left(\min(\mathrm{FRV}_j, C_j) - K_j\right). \tag{5}$$

If the cap is not applicable, the payoff to the volatility buyer is:

$$\Pi_j = A_j\left(\mathrm{FRV}_j - K_j\right). \tag{6}$$

The payoff to the volatility seller is the negative of the buyer payoff.

5 Dispersion Package Payoff

For a long-single-name-volatility and short-index-volatility dispersion package, the payoff is:

$$\Pi_{\text{disp}} = \sum_{j\in\mathcal{S}} A_j\left(\min(\mathrm{FRV}_j, C_j) - K_j\right) - A_I\left(\min(\mathrm{FRV}_I, C_I) - K_I\right). \tag{7}$$

This position benefits when realized single-name volatility is high relative to realized index volatility. Because index variance includes covariance terms, the package is closely related to a short correlation exposure:

$$\sigma_I^2 \approx \sum_j w_j^2 \sigma_j^2 + 2 \sum_{j<k} w_j w_k \rho_{jk} \sigma_j \sigma_k. \tag{8}$$

Lower realized correlation reduces index volatility relative to the basket of single-name volatilities, which is favorable for a long dispersion position.

6 Interim Valuation Before Maturity

At a calculation date before final maturity, suppose n log returns have already been observed. The realized contribution to annualized variance is:

$$\text{RealizedContribution}_j = \frac{252}{N} \sum_{i=1}^{n} r_{j,i}^2. \tag{9}$$

The important point is that this term is already divided by the full-period expected observation count N. It should not be multiplied again by n/N.

Let $K_{\text{var},j,\text{fwd}}^2$ be the forward variance rate for the remaining observation period. A practical projected variance estimate is:

$$\widehat{\sigma}_j^2 = \frac{252}{N} \sum_{i=1}^{n} r_{j,i}^2 + \frac{N-n}{N} K_{\text{var},j,\text{fwd}}^2. \tag{10}$$

The projected final realized volatility is then:

$$\widehat{\text{FRV}}_j = 100 \sqrt{\widehat{\sigma}_j^2}. \tag{11}$$

The interim mark-to-market approximation uses the same payoff function with $\widehat{\text{FRV}}_j$ replacing the unknown final FRV_j:

$$\widehat{\Pi}_{\text{disp}} = \sum_{j \in \mathcal{S}} A_j \left(\min(\widehat{\text{FRV}}_j, C_j) - K_j \right) - A_I \left(\min(\widehat{\text{FRV}}_I, C_I) - K_I \right). \tag{12}$$

7 Implied Variance Input

The theoretically cleaner input for the remaining period is a forward variance swap rate, not simply ATM implied volatility squared. The variance swap rate is extracted from a slice of the option surface at a fixed maturity T by integrating across strikes.

Let $F(T)$ be the forward price for expiry T, and let $Q(K,T)$ denote the undiscounted price of the out-of-the-money option with strike K and maturity T:

$$Q(K,T) = \begin{cases} P(K,T), & K < F(T), \\ C(K,T), & K > F(T), \end{cases} \tag{13}$$

where $P(K,T)$ is the put price and $C(K,T)$ is the call price. A continuous-strike approximation to the annualized variance swap rate is:

$$K_{\text{var}}^2(T) \approx \frac{2}{T}\left[\int_0^{F(T)} \frac{P(K,T)}{K^2}dK + \int_{F(T)}^{\infty} \frac{C(K,T)}{K^2}dK\right]. \tag{14}$$

Equivalently:

$$K_{\text{var}}^2(T) \approx \frac{2}{T}\int_0^{\infty} \frac{Q(K,T)}{K^2}dK. \tag{15}$$

The integral is across strikes on one fixed-expiry T slice of the volatility surface. The option surface has two dimensions, strike and maturity; this calculation freezes maturity at T and integrates only in the strike direction. The division by T converts the option strip's implied total variance over $[0,T]$ into an annualized variance rate.

In practice, the continuous integral is replaced by a discrete sum over listed strikes:

$$K_{\text{var}}^2(T) \approx \frac{2}{T}\sum_m \frac{\Delta K_m}{K_m^2}Q(K_m,T), \tag{16}$$

where ΔK_m is the strike interval around strike K_m. Market implementations also handle discounting, forwards, dividends, settlement conventions, bid-ask filtering, and far-wing extrapolation. The key point is that the full smile matters; a single ATM volatility does not capture skew or tail option prices.

For a remaining period from T_1 to T_2, use total implied variance and difference it:

$$K_{\text{var,fwd}}^2(T_1,T_2) = \frac{T_2K_{\text{var}}^2(T_2) - T_1K_{\text{var}}^2(T_1)}{T_2 - T_1}. \tag{17}$$

If the calculation date is today and the only remaining maturity is the final valuation date, then $T_1 = 0$ and the forward variance rate reduces to the variance swap rate for the final maturity slice:

$$K_{\text{var,fwd}}^2(0,T) = K_{\text{var}}^2(T). \tag{18}$$

If only ATM implied volatility is available, a rough proxy is:

$$K_{\text{var},j,\text{fwd}}^2 \approx \sigma_{\text{ATM},j,\text{fwd}}^2. \tag{19}$$

This proxy is acceptable for a quick sensitivity check, but it is not a true implied variance calculation. It will be most unreliable when the underlying has strong skew, wide wings, event risk, or sparse option strikes.